project.org

TEACHER LESSON PLAN

My Eyes Can Trick Me!

PreK–K · 20 minutes · General / Homeroom

A gentle introduction to optical illusions that teaches young students that eyes can sometimes "trick" us, and that it's okay to check and ask questions before deciding what we see.

THE 20 MINUTES

5 min	10 min	3 min	2 min
INTRODUCTION	WHAT DO YOU SEE?	REAL LIFE	CLOSING

LEARNING OBJECTIVE

Students will understand that their eyes and brain can sometimes see things differently than they actually are, and that it's okay to check and ask questions.

MATERIALS NEEDED

- Large printout of the duck/rabbit illusion
- Two identical colors of construction paper in different contexts (one surrounded by dark paper, one by light paper)
- **Mini Museum Perception Lab:** minimuseumproject.org/exhibitions/seeing-is-deceiving/illusions
- **Recommended from Perception Lab:** Rubin's Vase (#4 — faces or vase?)

INTRODUCTION · 5 MINUTES

Our eyes are amazing

"Today we're going to learn that our eyes are amazing, but sometimes they play tricks on us! Has anyone ever thought they saw something, but it turned out to be something different?"

Share a personal, age-appropriate story. Example: *"One time I thought I saw my friend across the street and I waved, but when I got closer, it was someone else who just looked like my friend from far away!"*

Introduce the duck/rabbit: Show the picture. *"Is this a duck or a rabbit?"* [Let students share what they see first.] *"It's both! Your brain can see it two different ways. That's called an optical illusion—when your eyes trick you."*

ACTIVITY · 10 MINUTES

"What Do You See?" Game

ROUND 1 · SIMPLE ILLUSIONS · 5 MINUTES

- Show 2–3 age-appropriate optical illusions
- For each: *"What do you see first? Can you see it a different way?"*
- Celebrate all answers: *"Great! Some people see it one way, some see it another way. Both are right!"*

ROUND 2 · COLOR MATCHING · 5 MINUTES

- Show two pieces of identical construction paper surrounded by different colored backgrounds
- *"Are these the same color or different colors?"* Take predictions
- Put them side by side to reveal they're identical

- *"See? What's around them changed how they looked! Your eyes got tricked by what was next to the colors."*

> *"It's okay if you weren't sure! Our eyes can trick us sometimes. That's why it's always good to look closer and check."*

CONNECTION TO REAL LIFE · 3 MINUTES

"Sometimes we might think a friend is mad at us because of how their face looks, but maybe they're just tired or thinking hard. Before we decide how someone feels, we should ask them! Just like checking if the colors are really the same."

Examples for young students:

- "If you think someone took your toy, check first—maybe it's just in a different spot"
- "If you think someone isn't being nice, ask them what they meant before getting upset"
- "If something looks scary from far away, get closer to see what it really is"

CLOSING · 2 MINUTES

"Remember: It's always okay to look closer and ask questions. Even grown-ups' eyes play tricks on them sometimes!"

Exit question: *"If your friend sees something different than you, what should you do?"* (Look again! / Ask questions! / Both can be right!)

> ### MINI MUSEUM CONNECTION
>
> **Option 1 — Visit the Mini Museum:** *"When you visit the Mini Museum, you'll see old toys that tricked people's eyes over 100 years ago! You'll see a special old camera called a stereoscope that makes flat pictures look 3D, and cards that change when you move them."*
>
> **Option 2 — Virtual Visit:** Show the exhibition online at minimuseumproject.org/exhibitions/seeing-is-deceiving or view individual artifacts at minimuseumproject.org/exhibitions/seeing-is-deceiving/artifacts

ASSESSMENT

Informal observation: During the activity, notice which students can identify that the images can be seen multiple ways, and which students grasp that "checking" is a good strategy.

Exit question: *"Can anyone tell me one thing you learned today about your eyes?"* Accept answers like "eyes can trick you," "check before you decide," "look closer," etc.

DIFFERENTIATED STUDENT HANDOUTS

Three levels of student handouts are provided as separate files. You can print whichever level(s) work for your students, or allow students to self-select their challenge level:

GREEN LEVEL MORE SUPPORT	YELLOW LEVEL GRADE LEVEL	RED LEVEL GO DEEPER
"I want more support" — Look at illusion pictures and circle what you see	"I'm ready for a challenge" — Draw your own trick picture	"I want to go deeper" — Create and test an illusion with others

These handouts can be used in class, sent home, or used as informal assessment. They include a "Share with Your Family" section to extend learning.

BOOK DISPLAY SUGGESTIONS

Create a "Look Again!" display featuring: *Duck! Rabbit!* by Amy Krouse Rosenthal, *Round Trip* by Ann Jonas, *Look! Look! Look!* by Tana Hoban, *It Looked Like Spilt Milk* by Charles G. Shaw, *I Spy* series, *Where's Waldo?*